

Nanostructure and poroviscoelasticity in cell wall materials from onion, carrot and apple: Roles of pectin

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ARTICLE INFO

Keywords:

Cell wall structure

Cellulose

Small angle x-ray scattering

Small angle neutron scattering

Rheology

ABSTRACT

The hierarchical organisation of polysaccharides in primary plant cell walls is responsible for their unique mechanical properties, and in turn for the textural and rheological properties of plant-based foods and ingredients. It is expected that at the nano scale, the mechanical properties of cell wall materials arise from a combination of structural deformation of the polysaccharide networks and hydraulic properties of the continuous water phase, as has been shown for other cellulose-based composites. Pectin plays a key role in the load bearing properties of (bacterial) cellulose-pectin composites due to its contribution to both hydration structure and the dynamics of water movement. To investigate whether these features are also important in plant cell wall materials we have used a set of advanced characterisation techniques to elucidate cell wall structural features at different length scales (X-ray diffraction and small angle X-ray and neutron scattering) in cell walls from two dicotyledons (apple and carrot) and a non-commelinid monocotyledon (onion). The strength of isolated cell walls was measured under compression and fitted to a poroviscoelastic mechanical model, demonstrating that the mechanical properties of the isolated cell wall materials are directly linked to both polysaccharide networks and fluid flow through the networks. Our results show how pectin polysaccharides influence the viscoelastic behaviour of these materials and contribute to the texture of plant-derived food systems.

1. Introduction

The properties of cell walls are of central importance to formulation and digestion of plant-based foods and ingredients from fruits and vegetables. However, structure/rheology relationship for plant tissues are not as well understood as for refined hydrocolloid ingredients. With the nutritional importance of foods based on fruits and vegetables and the opportunities for repurposing plant food wastes as food structuring agents, it is important that a more thorough understanding of the relationships between material properties of relevance to foods and both molecular composition and microstructure of cell wall materials is obtained.

The main polysaccharides of the primary cell wall of fruit and vegetable food materials consumed are cellulose, hemicelluloses and pectin polysaccharides. Pectins are complex block co-polymers and consist mainly of homogalacturonan (HG), rhamnogalacturonan I (RG

I) and rhamnogalacturonan II (RG II). Homogalacturonan (HG) is a negatively charged macromolecule that consists of α -1,4-linked galacturonic acid (GalA), often methyl-esterified at the C–O-6 carboxyl. In the presence of calcium ions, pectin is able to form calcium-pectate crosslinks, depending on the degree of methyl esterification (DE) and the distribution of ester groups. RGI and II are each highly branched polymers with the potential to occupy space within the polymer network affecting properties such as porosity (Mohnen, 2008). Pectin is a complex polymer whose chemical and molecular features make it versatile and therefore likely to have multiple functions *in muro*.

Although the compositions of common food plant cell walls are well documented, the microstructure of the cell wall is still not established, and models which were well accepted until recently are being challenged by new evidence resulting from *in vitro* studies on cell wall model systems (Lin, Lopez-Sanchez, & Gidley, 2015; Zykwiniska, Thibault, & Ralet, 2008) and *in vivo* studies on model plant materials

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<https://doi.org/10.1016/j.foodhyd.2019.105253>

Received 26 April 2019; Received in revised form 26 July 2019; Accepted 26 July 2019

Available online 26 July 2019

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such as Arabidopsis mutants (Cosgrove, 2014; Peaucelle, Braybrook, & Hofte, 2012), which indicate that pectin interacts directly with cellulose and contributes to cell wall mechanics. We have previously investigated the physicochemical principles of interactions between plant polysaccharides in cellulose composites of bacterial origin using small angle scattering techniques (Lopez-Sanchez et al., 2017; Martinez-Sanz, Lopez-Sanchez, Gidley, & Gilbert, 2015) as well as their role in mechanical and rheological properties (Lopez-Sanchez et al., 2015; Lopez-Sanchez et al., 2014). The use of advanced structural characterisation techniques, such as small angle x-ray and neutron scattering (SAXS and SANS), to plant materials is limited due to the complexity of plant materials and the inherent difficulty of interpreting scattering data on such complex systems. However, combined with complementary methods, they provide an efficient approach to characterise plant cell walls, covering a broad and relevant size range while minimising experimental artefacts derived from sample treatment (i.e. native plant tissues can be directly measured) (Martínez-Sanz, Gidley, & Gilbert, 2015).

Water flow in plant tissues/organs is responsible for the large stresses that plants are able to withstand due to their poroelastic nature i.e. the coupling of network architecture with fluid flow. Indeed, poroelastic principles have been shown to rationalise a broad range of plant tissue movements (Skotheim & Mahadevan, 2005). Skotheim et al. classified the physical rationale behind plant movements, such as explosive fractures to disperse seeds or the elastic buckling to catch insects of the Venus flytrap, in terms of mechanics of water transport through an elastic tissue. Furthermore, for cellulose hydrogel networks, a poroelastic model has been shown to be appropriate to describe mechanical properties (Lopez-Sanchez et al., 2014). In cellulose/pectin composites, a poroviscoelastic model is appropriate due to the viscoelastic property of pectins (Lopez-Sanchez et al., 2017). It is therefore of interest to investigate whether poro(visco)elasticity could also explain mechanics at the level of the plant cell wall and relevant to food properties such as texture and digestibility.

Our aim in this study was to understand relationships between the structural properties of cell wall materials and mechanical properties under compression, with a particular focus on the potential role of pectin. To this end, we relate the structural differences at the nanoscale with cell wall mechanical behaviour using a poroviscoelastic model for pectin-rich primary cell walls from apple, carrot and onion.

2. Materials and methods

2.1. Cell wall preparation

Carrot (*Daucus carota*), apple (*Malus pumila*) and onion (*Allium cepa*), were selected as representative of well-characterised plant materials. Cell wall materials were prepared following the methods of Pettolino et al. (Pettolino, Walsh, Fincher, & Bacic, 2012) and Bindon et al. (Bindon, Smith, & Kennedy, 2010) with some modifications. Edible parts of carrot root, apple fruit and onion bulb were peeled (the core of the apples was discarded) and cut into small pieces of approximately 1 cm³. Pieces were ground in the presence of ice-cold 40 mM HEPES buffer (pH 7) to solubilise most cell contents. After that the material was centrifuged (10000 g for 20 min at 4 °C to prevent action of wall enzymes), and the sediment was collected and mixed again with HEPES buffer (400 mL) and further ground using an ultraturax at 20000 rpm for 1 min. A second centrifugation step was performed (10000 g for 20 min at 4 °C) and the supernatant discarded. The alcohol-insoluble residue (AIR) was prepared from the HEPES-treated cell wall material by washing twice with 80% v/v ethanol for 30 min on ice and centrifuged (10000 g for 5 min at 4 °C) followed by washing twice in 70% v/v acetone at room temperature and acetone-insoluble residues were then washed for 30 min in 1:1 methanol: chloroform with gentle mixing. Iodine/KI stain was used to confirm that prepared cell wall materials did not contain starch. Cell walls were finally washed

with deionised water to remove excess acetone which could interfere with the scattering experiments. This preparation method ensures preservation of the native cell wall scaffold whilst separating all water-soluble cell wall components and cell contents.

The structural and mechanical roles of pectin were further investigated in onion cell walls by extracting acid-soluble (acid extracted) and chelator-soluble pectin (EDTA extracted). Acid-soluble pectins were removed by mixing the HEPES extracted cell walls in 1% w:v citric acid pH 2.3 (1:6) whilst stirring at 500 rpm at 50 °C for 1 h. Samples were cooled in an ice bath and centrifuged at 10000 g for 20 min. The sediment was collected and mixed with deionised water (1:1) whilst stirring at 500 rpm at 50 °C for 30 min and centrifuged again at 10000 g for 20 min (Yuliarti, Goh, Matia-Merino, Mawson, & Brennan, 2015). The sediment was collected and vacuum filtered whilst washing with 400 mL of deionised water. Chelator-soluble pectins were removed by treating the acid extracted cell walls with 40 mM Na₂EDTA (1:20), the pH was adjusted to 3.3 with 6 M and 0.1 M HCl and the mixture was stirred at 500 rpm at 85 °C for 1 h. Samples were centrifuged and the sediment collected by vacuum filtration whilst washing with 400 mL of deionised water.

The moisture content of each cell wall extract was measured by drying the samples in an oven at 105 °C for 24 h. The sample weight was measured before and after drying. Each sample was measured in duplicate.

Cell wall suspensions were adjusted to a concentration of 7% w:v by gently mixing the cell wall extract with the appropriate amount of deionised water. This concentration was based on the wet weight obtained from the extracts. Suspensions were allowed to stabilise for 24 h prior to analysis.

2.2. Monosaccharide analysis

Cell wall monosaccharide composition was determined using the method of Pettolino et al., (Pettolino, Walsh, Fincher, & Bacic, 2012) involving derivatization to alditol acetates after initial hydrolysis and reduction. Freeze dried cell wall materials (5 mg) were hydrolysed at room temperature for 1 h using 12 M H₂SO₄. Myo-Inositol was added as an internal standard. Samples were reduced using 20 mg/mL sodium borodeuteride in dimethylsulphoxide (DMSO) and acetylated by adding 1-methylimidazole followed by acetic anhydride. Dichloromethane (DCM) was used to extract the alditol acetates. Monosaccharide content was analysed by GC-MS (QP2010 Ultra, Shimadzu) using a high polarity column (BPX70, SGE, 25 m × 0.32 mm inner diameter, film = 0.25 µm) and calculated for individual sugar contents on the basis of percentage dry weight.

2.3. Uronic acid assay

Total uronic acid content was measured according to a previously reported colorimetric method (Filisetti-Cozzi & Carpita, 1991) with slight modifications. Freeze dried samples (3 mg) were hydrolysed in 1.0 mL of chilled concentrated H₂SO₄ (98%) for 10 min, then diluted to 10 mL using distilled H₂O. D-Galacturonic acid solutions (0–200 µg/mL) in distilled H₂O were also prepared as calibration standards. Aliquots of hydrolysed samples and standards (160 µL) were mixed with 4 M sulfamic acid-potassium sulfamate solution (20 µL in KOH, pH 1.6) and the resulting mixtures were diluted using sodium tetraborate (12.5 mM, 800 µL) dissolved in concentrated H₂SO₄. These were incubated at 95 °C for 20 min, then cooled over ice. The subsequent maximum change in absorbance at 525 nm was measured using a spectrophotometer (UV-1700, Shimadzu) upon addition of 3-phenylphenol (40 µL, 0.15% w/v). Total uronic acid content was calculated as the percentage dry weight.

2.4. X-ray diffraction (XRD) experiments

The crystalline structure of cell walls was investigated by XRD

measurements according to the method described by Martínez-Sanz et al. (Martínez-Sanz, Pettolino, Flanagan, Gidley, & Gilbert, 2017). Experiments were carried out in a PANalytical X'Pert Pro diffractometer and peak fitting was carried out as described in (Martínez-Sanz et al., 2017).

2.5. Small angle X-ray and neutron scattering experiments (SAXS and SANS)

SAXS measurements were performed following the method described by Martínez-Sanz et al. (Martínez-Sanz et al., 2015; Martínez-Sanz et al., 2017) on a Bruker Nanostar with a rotating anode source operated at wavelength, λ , of 1.54 Å (Cu K α) with a sample-to-detector distance of 723 mm, giving a q range from 0.01 to 0.39 Å⁻¹. The exposure time was 5400 s for the hydrated samples and 3600 s for the air-dried samples and backgrounds. All the samples were measured with one single exposure. SANS measurements were performed on the 40 m QUOKKA instrument at the OPAL reactor (Gilbert, Schulz, & Noakes, 2006; Wood et al., 2018), as described previously (Martínez-Sanz et al., 2015). The SANS and SAXS data were fitted by applying the core-shell model described in previous reports (Martínez-Sanz, Mikkelsen, Flanagan, Gidley, & Gilbert, 2016; Martínez-Sanz et al., 2017).

2.6. Compression-stress relaxation experiments

Compression and stress-relaxation experiments were carried out on a stress-controlled rheometer (HAAKE Mars III Rheometer, Thermo Fisher Scientific, Karlsruhe, Germany) with a heat adjustable Peltier element at a temperature of 25 °C, and an attached 35 mm diameter plate coated with sand paper to reduce sample slip during measurements. Cell wall materials were loaded on the rheometer and the gap between the plates was adjusted to 1.5 mm. Cell wall materials were compressed to a gap of 100 μ m at a constant speed of 1 μ m/s while measuring the normal force (Lopez-Sanchez et al., 2014). After the compression step, a small amplitude oscillatory shear (SAOS) deformation test was performed for 300 s at a frequency of 0.016 rad/s (1 Hz) and at a low constant stress of 1 Pa, chosen within the linear viscoelastic region. G' and G'' (see Supplementary Material Fig. S3); values at time 300 s were selected for comparison. Normal force was recorded during the SAOS test to follow stress relaxation. SAOS experiments allow for investigations to be performed under unperturbed conditions and determines viscoelastic moduli G' and G'' . Samples were measured in triplicate.

2.7. Poroviscoelastic modelling

The cell wall samples were modelled as a poroviscoelastic system. Poroviscoelasticity is the term used to describe the interaction between fluid flow and solids deformation within a porous medium. Intrinsic viscoelasticity arises due to the relaxation of the constituent polymer chains and, along with poroviscoelasticity, is a fundamental component of

the time-dependant response to compression of cellulose composites (Lopez-Sanchez et al., 2017). For simplicity, we consider a linear isotropic poroviscoelastic material, following the prescription of DiSilvestro et al. (DiSilvestro, Zhu, Wong, Jurvelin, & Suh, 2001). While the cell wall *in planta* must be anisotropic to perform its function (Baskin, 2005), the samples in this work are essentially isotropic, as they consist of suspended cell wall fragments.

The governing equations for a poroviscoelastic system are:

$$\text{Continuity equation: } \nabla \cdot (\sigma^s \mathbf{v}^s + \sigma^f \mathbf{v}^f) = 0 \quad (1)$$

$$\text{Momentum equation: } \nabla \cdot \sigma^\alpha + \pi^\alpha = 0; \quad \alpha = s, f \quad (2)$$

$$\text{Constitutive equations: } \sigma^f = \varphi^f p \mathbf{I} \quad (3)$$

$$\sigma^s = \varphi^s p \mathbf{I} + \tilde{\sigma}^s \quad (4)$$

$$\pi^s = \pi^f = \frac{\eta(\varphi^f)^2}{\kappa} (\mathbf{v}^f - \mathbf{v}^s) \quad (5)$$

where σ is the stress tensor, $\mathbf{v}$ is the velocity vector, π is the diffusive body force vector, φ is volume fraction, κ is permeability, η is fluid viscosity and p is interstitial fluid pressure. The superscripts s or f denote the solid or fluid phase, respectively. $\tilde{\sigma}^s$ is the effective solid stress tensor representing the portion of the solid stress that is in excess of the local fluid pressure. The linear poroviscoelastic model accounts for intrinsic viscoelasticity of the solid phase, in this case cell wall material, independent of bulk flow of the interstitial fluid. To do this, the effective solid stress, $\tilde{\sigma}^s$ is adjusted to account for the elasticity of the solid phase in bulk deformation and the viscoelasticity of the solid phase in deviatoric deformation (DiSilvestro et al., 2001). For a linear viscoelastic system, $\tilde{\sigma}^s$ becomes,

$$\tilde{\sigma}^s = -\frac{E}{3(1-2\nu)} \mathbf{I} + \frac{E}{2(1+\nu)} \int_0^t G(t-\tau) \frac{\partial \mathbf{e}^s}{\partial \tau} d\tau \quad (6)$$

$$G(t) = 1 + \bar{G} e^{-t/\tau_v} \quad (7)$$

where E is the Young's modulus, ν the Poisson's ratio, $\mathbf{e}^s$ is the deviatoric component of the solid strain tensor (i.e. $\mathbf{e}^s = \boldsymbol{\varepsilon}^s - 1/3 \text{tr}(\boldsymbol{\varepsilon}^s) \mathbf{I}$), $\bar{G}$ is the magnitude of the relaxation spectrum and τ_v the viscoelastic relaxation time.

Hence, the fitting parameters are E , ν , k , $\bar{G}$ and τ_v . The poroelastic time τ_p , representing the timescale for water transport in a continuum of soft porous media is related to the system properties and the radius R of the sample through $\tau_p = \eta R^2 / kE$. The system of equation (1) – (7) was solved using a finite elements implementation in Matlab, assuming (i) unconfined compression, and (ii) homogenous impermeable contact with the rheometer surfaces. Further details on the experimental approach can be found elsewhere (Lopez-Sanchez et al., 2014).

Table 1

Monosaccharide composition and total uronic acids (calculated as mol %). Cell walls from apple, carrot and onion were extracted using HEPES buffer (40 mM). Onion cell walls were further investigated by extracting acid-soluble (acid extracted) and chelator-soluble pectin (EDTA extracted). Values show the average of two replicates with standard deviation. n.d. = not detected.

	Apple HEPES extracted	Carrot HEPES extracted	Onion HEPES extracted	Onion acid extracted	Onion EDTA extracted
Rha	1.2 $\pm$ 0.1	1.2 $\pm$ 0.2	1.2 $\pm$ 0.2	1.2 $\pm$ 0.1	0.8 $\pm$ 0.2
Fuc	n.d.	n.d.	n.d.	n.d.	n.d.
Ara	7.6 $\pm$ 1.7	7.9 $\pm$ 0.1	0.9 $\pm$ 0.1	1.1 $\pm$ 0.2	1.5 $\pm$ 0.6
Xyl	6.6 $\pm$ 1.1	0.4 $\pm$ 0.1	1.7 $\pm$ 0.1	2.4 $\pm$ 0.0	2.8 $\pm$ 0.1
Man	1.1 $\pm$ 0.1	1.2 $\pm$ 0.0	0.8 $\pm$ 0.0	0.6 $\pm$ 0.4	1.3 $\pm$ 0.1
Gal	4.6 $\pm$ 0.7	13.3 $\pm$ 0.0	15.9 $\pm$ 0.2	18.8 $\pm$ 0.3	17.3 $\pm$ 0.1
Glc	43.5 $\pm$ 1.8	38.8 $\pm$ 0.3	26.5 $\pm$ 0.1	32.1 $\pm$ 0.2	44.8 $\pm$ 0.5
Uronic acids	35.4 $\pm$ 0.1	37.2 $\pm$ 0.01	53.0 $\pm$ 0.01	43.8 $\pm$ 0.01	31.5 $\pm$ 0.02

3. Results and discussion

3.1. Chemical composition and crystallinity of cell walls

The main monosaccharides measured in the cell wall preparations are shown in Table 1. Glucose contents of 43.5 mol %, 38.8 mol % and 26.5 mol % were measured in apple, carrot and onion respectively. Onion cell walls were low in arabinose, 0.9 mol %, compared to 7.6% in apple and 7.9% in carrot cell walls. Xylose content was higher in apple and the galactose content varied among the three materials: 4.6 mol % for apple, 13.3% for carrot and 15.9% for onion; galactose in these primary cell walls is mostly present as galactan, a main sidechain component of rhamnogalacturonan I pectin. The content of uronic acids was higher in onion cell walls (53.0 mol %) compared to apple (35.4%) and carrot (37.2%). Similar compositions have been previously reported for these plant materials (Houben, Jolie, Fraeye, Van Loey, & Hendrickx, 2011; Ng, Smith, & Waldron, 1998; Videcoq et al., 2017). Acid-extracted onion cell walls showed a reduction in uronic acid content to 43.8 mol %. Cell walls which were treated with EDTA, as a calcium chelator, showed a further reduction in total uronic acids to 31.5%, and slightly different values for other monosaccharides (e.g. galactose, 17.3%, rhamnose 0.8%), confirming that different pectins were extracted as a result of the different treatments.

Cell wall crystallinity influences the impact of food manufacturing conditions e.g. thermal and mechanical treatments, as well as food digestion of plant materials e.g. access of enzymes to their substrates. A relatively low crystallinity index (35–55%), compared to more cellulose-rich plant cell wall sources such as cotton ($X_C \sim 63\text{--}66\%$ (Martínez-Sanz et al., 2017)) or ramie ($X_C \sim 83\%$ (Revol, Dietrich, & Goring, 1987)), was measured for all three species (Table 2). Cell walls presented diffraction patterns characteristic of cellulose I_β allomorph, with the (1-10) and (110) peaks overlapped (Fig. 1), similar to cotton cell wall materials (Martínez-Sanz et al., 2017). A large variability is observed for the crystallinity values reported in the literature for these sources, which is most likely due to the different processes used to isolate the cell walls and the different techniques and approaches utilised to calculate the crystallinity values. For instance, a value of ca. 25% has been estimated by means of Raman spectroscopy for the crystallinity of cellulose extracted from apples (Szymańska-Chargot, Cybulska, & Zdunek, 2011), whereas values of ca. 12–16% (Georget, Cairns, Smith, & Waldron, 1999) and 21–56% (Rhim, Reddy, & Luo, 2015; Shimazu & Sterling, 1967; Shimazu, Sterling, & York, 1965) have been reported for carrot and onion cell walls, respectively, by means of XRD analyses.

The shape of the diffraction peaks was markedly different (Fig. 1) in onion cell walls compared to apple and carrot cell walls. The XRD pattern from native onion cell walls is similar to that previously observed when phase separation occurs between cellulose and pectin in the absence of calcium in bacterial cellulose-pectin composites (Lopez-Sanchez et al., 2016), despite the differences in cellulose dimensions and crystallinity of those composites. Regarding the crystallite size, similar dimensions of 2.7×2.4 nm have been previously calculated

Table 2

Apparent crystallinity index (X_C) and cross-sectional dimensions of crystallite sizes in the direction perpendicular to the (1-10), (110) and (200) planes ($D_{(1-10)}$, $D_{(110)}$ and $D_{(200)}$), determined from the XRD patterns of air-dried samples.

	Apple HEPES extracted	Carrot HEPES extracted	Onion HEPES extracted	Onion acid extracted	Onion EDTA extracted
X_C (%)	38 ± 3	38 ± 1	37 ± 2	55 ± 2	35 ± 9
$D_{(1-10)}$ (nm)	2 ± 5	3 ± 2	3 ± 1	4 ± 2	4 ± 3
$D_{(110)}$ (nm)	2 ± 1	6 ± 9	3 ± 3	2 ± 9	3 ± 2
$D_{(200)}$ (nm)	3 ± 3	3 ± 3	3 ± 5	3 ± 4	3 ± 3

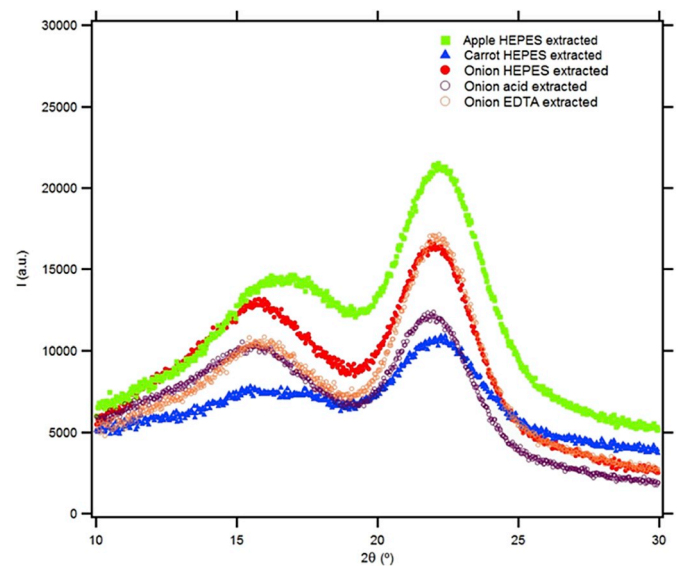


Fig. 1. XRD patterns of air-dried cell walls from apple, carrot and onion.

from the ^{13}C NMR characterization of onion cell walls (Smith, Harris, Melton, & Newman, 1998). Crystallinity was higher in acid-extracted onion cell walls; however, it decreased again to values similar to the HEPES extracted cell walls after EDTA treatment. These changes in crystallinity index could be explained by the different pectin fractions which are extracted by acid and EDTA. The signal from loosely bound pectin, located coating the cellulose microfibrils, could interfere with the signal from the cellulose, leading to estimated apparent crystallinity values lower than the actual crystallinity of the cellulose microfibrils. After the acid treatment the loosely bound pectins, i.e. those which undergo phase separation and affect the shape of the diffraction peaks, are removed thus resulting in higher apparent crystallinity values. However, after the EDTA treatment, part of the previously self-cross-linked pectin is released and able to interfere again with the signal from the cellulose, leading to the calculation of lower crystallinity values (despite the crystalline structure from the cellulose remaining the same).

3.2. Nanostructure of cell walls investigated by SAXS and SANS

Small-angle X-ray and neutron scattering (SAXS and SANS) were used to characterise the architecture of cellulose microfibrils in the different plant cell wall materials. As observed in Fig. 2A, the SAXS patterns from the H_2O -hydrated HEPES extracted cell wall materials are characterized by the appearance of a very broad shoulder feature, similar to that observed in other plant cell wall systems such as cotton (Martínez-Sanz et al., 2017). The appearance of this shoulder has been related to the core-shell structure of cellulose microfibrils and the ability of water to penetrate the space between the microfibrils (Martínez-Sanz et al., 2016; Martínez-Sanz et al., 2017). If the cellulose microfibrils are considered to consist of a core containing impermeable cellulose crystallites and a shell containing hydrated paracrystalline cellulose, a difference in scattering length density (SLD) generates a contrast between these two regions, as well as with the surrounding solvent. Based on this, a core-shell fitting function has been previously developed and demonstrated to fit the SAXS data from hydrated cellulose from different origins (Martínez-Sanz et al., 2015; Martínez-Sanz et al., 2016; Martínez-Sanz et al., 2017). Application of this fitting model provided satisfactory fits for the experimental data from the hydrated HEPES extracted cell walls (the reduced χ^2 value associated to the fitting results was 3.62), as shown in Fig. 2A. As deduced from the obtained fitting parameters, summarized in Table S1, onion had the smallest microfibril cross-section (ca. 2.1 nm) with the smallest relative

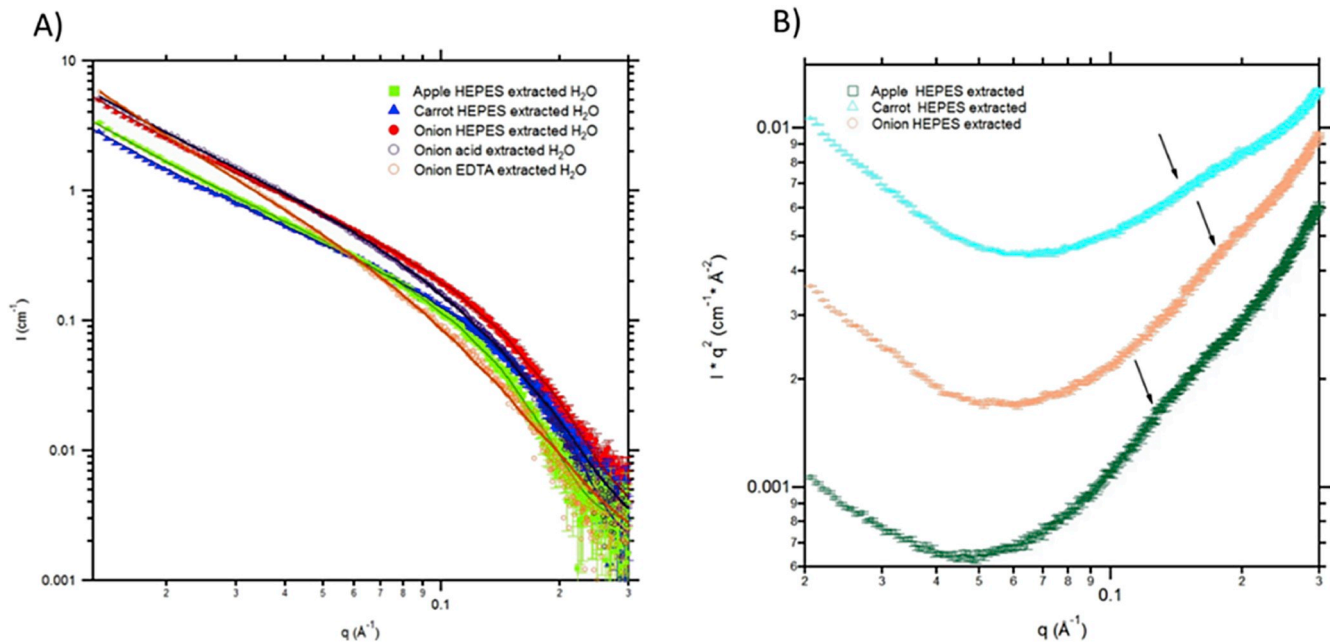


Fig. 2. (A) SAXS patterns of the HEPES extracted samples hydrated in H_2O . Solid lines correspond to the fitting of the experimental data using the core-shell model. (B) SAXS Kratky plot of air-dried HEPES extracted samples.

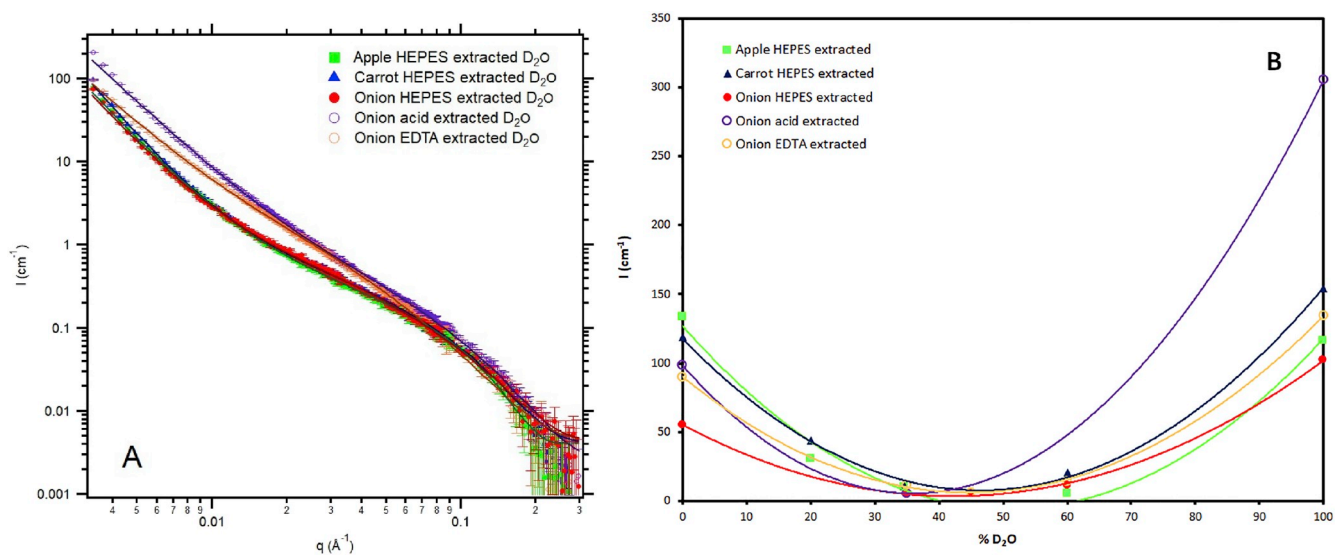


Fig. 3. (A) SANS data for samples soaked in D_2O . Solid lines correspond to the fitting of the experimental data using the core-shell model. (B) Contrast variation curves obtained from the SANS experiments using different H_2O/D_2O mixtures. Markers represent the experimental values and solid lines correspond to the fitted parabolic functions.

shell thickness, greatest core radius polydispersity and cellulose volume fraction. This seems to be due to the amount of pectin in the sample since the more pectin present in the material, the smaller is the calculated microfibril cross-section in the characterised samples (Table 1). Moreover, the microfibril cross-sections are found to increase when partially removing pectin from the onion cell walls, yielding an overall cross-section of ca. 3.4 nm for the acid extracted and ca. 4.0 nm for the EDTA extracted onion cell walls. Other studies have also found evidence that the pectin content affects cellulose microfibril dimensions (Smith et al., 1998).

In the air-dried state, the shoulder features became more defined (due to the increased SLD contrast between the microfibrils and the surrounding air) and appeared as weak peaks centred at ca. $0.16 \text{ \AA}^{-1}$ for onion, $0.14 \text{ \AA}^{-1}$ for carrot and $0.13 \text{ \AA}^{-1}$ for apple. The peak position was determined by fitting with a power-law plus Lorentzian peak

function, corresponding to distances of ca. 3.8 nm, 4.4 nm and 5.0 nm, respectively. This peak has been previously assigned to the centre-to-centre distance between individual cellulose microfibrils, which are relatively closely packed together in the dry state, and in some regions, along the microfibril longitudinal axis (Kennedy, Cameron, et al., 2007; Kennedy, Štuncová, Jarvis, & Wess, 2007; Martínez-Sanz et al., 2015; Martínez-Sanz et al., 2017). It is worth noting that the interfibrillar peaks observed for the HEPES extracted cell walls were not as defined as those detected for other cellulose sources, such as bacterial cellulose or cotton (sources which are almost pure cellulose), (Martínez-Sanz et al., 2015; Martínez-Sanz et al., 2016; Martínez-Sanz et al., 2017). This observation is consistent with the hypothesis, previously drawn from the characterisation of bacterial cellulose hydrogels synthesised in the presence of pectins (Lopez-Sanchez et al., 2017; Lopez-Sanchez et al., 2016), that the presence of matrix polysaccharides, such as pectic

polysaccharides, leads to the formation of more loosely packed cellulose microfibrils, even in the dry state.

The SAXS results indicate that when hydrating the cell wall samples in H₂O, moisture is able to penetrate the space between the microfibrils. Similar structural features, due to the SLD contrast generated between the core-shell microfibrils and the surrounding solvent, would therefore also be expected to appear in the SANS patterns. To corroborate this and to investigate the extent of H/D exchange undergone by cellulose upon contact with D₂O, contrast variation experiments were carried out by soaking the cell wall samples in different H₂O/D₂O mixtures.

As observed in Fig. 3A, the SANS patterns showed shoulder features similar to those detected in SAXS, which became less evident for the samples soaked in 35–60% D₂O. The absence of shoulder-like features within the q region of ca. 0.01–0.05 Å⁻¹, as reported for pure bacterial cellulose hydrogels (Martínez-Sanz et al., 2015; Martínez-Sanz et al., 2016), indicates that D₂O (or H₂O) is able to penetrate the space between the microfibrils. Therefore, the SLD contrast is between the impermeable microfibril cores and the surrounding hydrated shells, which apart from cellulose, may also contain cell wall matrix components such as pectic polysaccharides and xyloglucan. The same behaviour was observed for cotton samples (Martínez-Sanz et al., 2017), suggesting that the additional components present in plant cell walls during cellulose biosynthesis prevent the creation of the strongly hydrogen-bonded cellulose network formed in the absence of those components, as found in e.g. bacterial cellulose.

The results from the SANS contrast variation experiments (Fig. S1) were fitted using the same core-shell model used for SAXS (the reduced χ^2 values associated to the fitting results were 3.15 for onion CWs, 2.63 for carrot CWs, and 2.93 for apple CWs). The fitting results, summarized in Table S2 and schematically represented in Fig. S2, confirm that onion has the smallest microfibril dimensions. Furthermore, the cellulose volume fraction is ca. 0.8 for onion while it is ca. 0.4 for carrot and apple, suggesting that pectin has a densification effect, similar to that observed in bacterial cellulose-pectin composites, causing the amount of water held within the cellulose paracrystalline fraction to be reduced (Lopez-Sanchez et al., 2017). This is further supported by the decreased solvent H/D exchange within the microfibril shell estimated for the onion samples (ca. 20% compared to ca. 80% for carrot and ca. 90% for apple).

The SLD values from the cell wall samples were estimated from the contrast variation experiments. The scattering intensity at a q value of 0.0040 Å⁻¹ was plotted against the D₂O content in the solvent mixtures and the resultant curve was fitted to a parabolic function as shown in Fig. 3B. The contrast match point was then determined as the value at which the intensity reached its minimum. In a single-phase system, the SLD from the sample equals that of the solvent at the contrast match point; the SLD values for the different plant cell wall materials were estimated and the results are shown in Table 3. The first thing to note is that the calculated SLD values are greater than that of pure cellulose ($1.87 \times 10^{10} \text{ cm}^{-2}$), confirming the presence of additional components in the materials. It has been previously demonstrated that the presence of cell wall components, such as xyloglucan, may promote H/D exchange and therefore, affect the SLD values from cellulose microfibrils (Martínez-Sanz et al., 2016). The lower SLD value for onion is in agreement with the reduced degree of solvent exchange estimated from

the SANS data fitting and suggests that the relatively high level of pectin in onion leads to a microfibril densification effect and limits the solvent accessibility. This effect is similar to that observed in simplified model bacterial cellulose composites synthesised in the presence of Capectate gels (Lopez-Sanchez et al., 2017). We propose that calcium cross-linking of pectin limits pectin-cellulose interactions and thus reduces the microfibril densification effect of pectin.

After the sequential extraction of some of the pectin from onion cell walls, the SLD of the system was significantly affected. As observed in Table 3, the SLD decreased with acid extraction (suggesting limited H/D exchange) but increased again after EDTA extraction. We propose that only those pectins interacting more strongly with cellulose remain after the acid treatment. These pectins have a microfibril densification effect and limit H/D exchange. The EDTA treatment seems to have a structural effect on this fraction of pectins; they may become more loosely bound to cellulose after the treatment and thus, far from limiting H/D exchange, they instead provide additional labile OH groups for the exchange process to take place.

3.3. Compression and stress relaxation of cell wall materials

Compression of cell wall materials showed typical viscoelastic behaviour with a rapid increase in normal stress at low strains followed by a slower increase at higher strains. During compression both structural deformation and dynamics of fluid flow contribute to the mechanical behaviour. In Fig. 4a normal stress versus time curves are plotted. Onion cell walls showed higher normal stress of $23 \pm 0.2 \text{ kPa}$ than those from apple and carrot, which presented similar normal stresses of ca. 2.5 kPa (Table 4). All cell wall materials contained the same amount of total solids (7 % wt); however, their composition differed as shown in Table 1. Onion contained larger amounts of galacturonic acid compared to carrot and apple.

Cell walls, which is related to higher pectin content; indeed, it has been previously reported that onion cell walls are rich in pectins (Ng et al., 1998). Furthermore, scattering experiments demonstrated that onion microfibrils had the narrowest cross section and highest cellulose volume fraction, leading to denser microfibrils in which solvent exchange was more difficult (Table 3). We propose that the pectin present in onion cell walls is responsible for the high resistance to compression due to hydrated pectin occupying space between cellulose fibrils, which leads to a decrease in the ability of fluid from within cell walls to flow during compression. This leads to a higher pressure generated by the fluid, which thus contributes to an increase in the normal stress.

Fig. 4b depicts the relative compressive stress (i.e., the ratio between compressive stress at time t and the peak compressive stress) for the HEPES extracted walls shown in Fig. 4a. Normalizing the curves allows one to visualise the variations in viscoelastic recovery without the misleading effect of differences in stress magnitude. Viscoelasticity is produced by water retention within the pore structure of the pectin matrix (poroelasticity) and the intrinsic viscoelasticity of the polymer network. The corresponding poroelastic (τ_p) and intrinsic viscoelastic (τ_v) times obtained from fitting the experimental data are reported in Table 4. Apple cell walls are the quickest to reach the equilibrium stress during the relaxation stage, with onion and carrot cell walls displaying similar stress-decay rates during relaxation.

However, the model indicates that it will ultimately take more time for the onion cell wall to reach equilibrium, as a consequence of its higher poroelastic time. There is a faster viscoelastic decay for apple than both onion and carrot at the beginning of the relaxation stage (between 100 and ~130s) that may be related to the relative levels of pectin sidechains (Table 1). Apple has much less galactose (expected to be predominantly pectic galactan sidechains) than carrot and onion. Consistent with the suggested role for pectic galactan in influencing cell wall mechanics (Bidhendi & Geitmann, 2016; Ulvskov et al., 2005) we propose that this leads to a more rapid initial relaxation for apple cell walls because there is less galactan available to impede fluid flow

Table 3

Contrast match point and neutron scattering length density (SLD) values estimated from SANS contrast variation experiments.

	Contrast match (%D ₂ O)	SLD ($\cdot 10^{10} \text{ cm}^{-2}$)
Apple HEPES extracted	51 $\pm$ 1	2.97 $\pm$ 0.04
Carrot HEPES extracted	46.4 $\pm$ 0.2	2.66 $\pm$ 0.01
Onion HEPES extracted	42.1 $\pm$ 0.7	2.36 $\pm$ 0.02
Onion acid extracted	35.8 $\pm$ 0.1	1.92 $\pm$ 0.01
Onion EDTA extracted	44.8 $\pm$ 0.5	2.55 $\pm$ 0.02

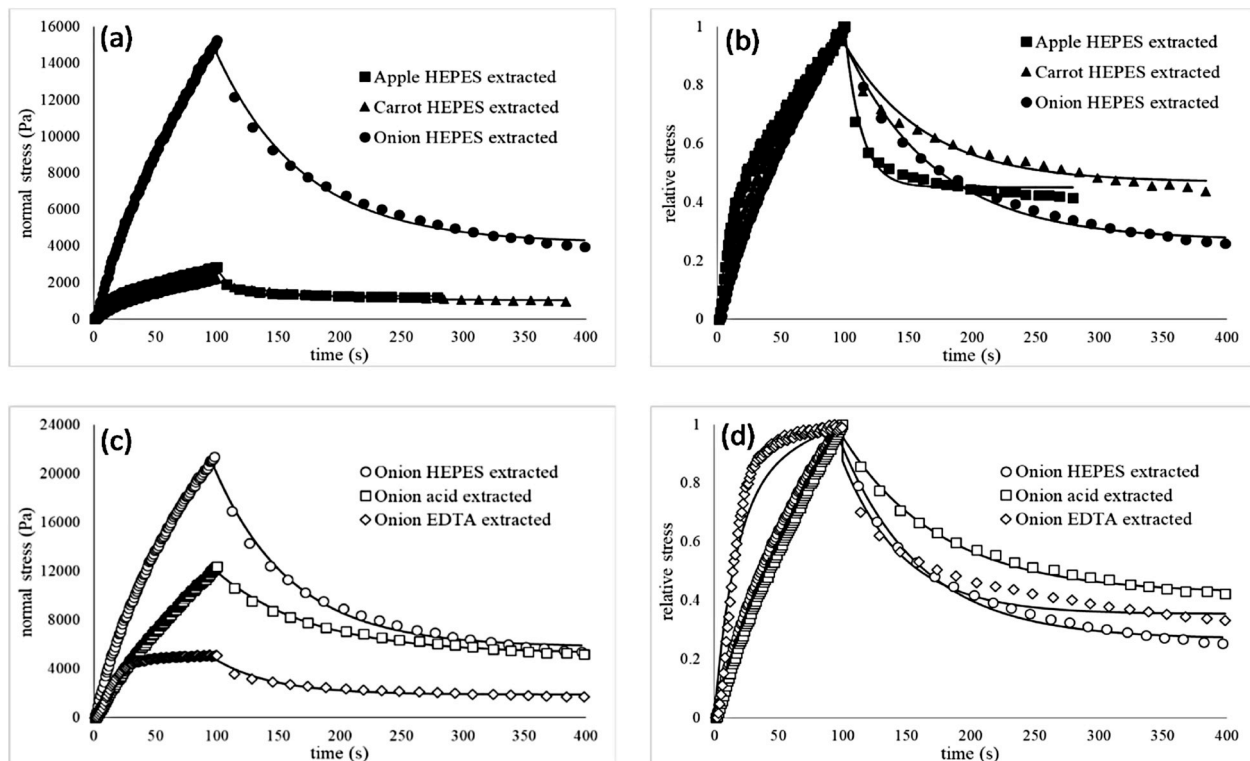


Fig. 4. Compression-relaxation curves of 7% w/w cell walls from (a) HEPES extracted carrots, apple and onion cell walls and, (c) onion cell walls after sequential extraction in HEPES, citric acid and Na₂EDTA. (b) relative stress (stress/peak stress) for the data in figure (a) and, (d) relative stress (stress/peak stress) for the data in figure (c). Symbols represent experiment and solid lines represent fitting to the poroelastic model.

through pores in the cellulose network. Interestingly, these poroviscoelastic times are in good agreement with viscoelastic times for 1.5% high methoxyl pectin gels (Mitchell, 1980), further suggesting the relevance of pectin in cell wall viscoelasticity control. Finally, it is important to clarify that neither onion nor carrot cell walls reached equilibrium during the experimental time (Fig. 4), unlike cellulose networks (Lopez-Sanchez et al., 2014). This means that additional, very slow pectin-driven viscoelastic effects could be occurring, on a longer timescale than used in the current experiments.

To further investigate the role of pectin, onion cell walls were treated with acid and EDTA to partially extract acid-soluble and chelator-soluble fractions respectively. After those treatments a fraction of pectin still remains (33 % wt.). Fig. 4c shows that the normal stress, indicative of resistance to compression of the cell walls, decreased when different pectin fractions were extracted. These results suggest that pectin contributes to the load bearing ability of the cell walls even when no covalent bonds exist between pectin and other cell wall components such as cellulose. It has been previously shown that the presence of a viscous pectin polymer influences the mechanical properties of

cellulose composites through poroviscoelastic effects (Lopez-Sanchez et al., 2016); furthermore, calcium-pectin networks contribute not only to the mechanics but also to the viscoelasticity of cellulose composites (Lopez-Sanchez et al., 2017). The relaxation behaviour of the cell walls shows that they did not relax completely during the time of the experiment as reflected by a residual normal stress.

Fig. 4d shows the relative compressive stress for the onion cell walls in Fig. 4c. Notably, there is a small reduction in poroviscoelastic times from HEPES extracted to acid-treated cell wall (see Table 4). This reduction has an important effect on the compressive stress, being partially responsible for the decrease in peak stress occurring upon removal of acid soluble pectin. On the other hand, extracting chelator-soluble pectin has a significant impact on the poroviscoelastic times, suggesting that (i) a non-negligible fraction of water is retained by the chelator-soluble pectin fraction and (ii) this fraction, which interacts with cellulose, could be effectively linking individual microfibrils and damping relaxation of the cellulose network.

Table 4 shows that apart from modulating the walls' viscoelasticity, pectin has an important contribution to the axial moduli. More

Table 4

Peak compression stress and viscoelastic moduli G' and G'' obtained experimentally for cell wall material with a concentration of 7% (wet weight). Different letters in the same column denote significant differences. All other parameters were obtained using the linear isotropic poroviscoelastic model with a relative fitting error of 12–15%.

	Apple HEPES extracted	Carrot HEPES extracted	Onion HEPES extracted	Onion acid extracted	Onion EDTA extracted
Experimental peak compression stress (kPa)	2.7 ± 0.2 ^c	2.5 ± 0.6 ^c	23 ± 0.2 ^a	13 ± 1.5 ^b	4.3 ± 0.5 ^c
G' (kPa)	26 ± 4 ^c	32 ± 8 ^c	101 ± 5 ^a	60 ± 10 ^b	25 ± 1 ^c
G'' (kPa)	3.7 ± 0.6 ^c	6.0 ± 2.4 ^c	34 ± 0.9 ^a	15 ± 3 ^b	6 ± 0.2 ^c
Tan δ	0.2 ± 0.3 ^a	0.1 ± 0.2 ^a	0.3 ± 0.2 ^a	0.2 ± 0.3 ^a	0.2 ± 0.2 ^a
E (kPa)	19.7	24.8	202	144	34
ν	0.28	0.33	0.40	0.32	0.34
τ_v (s)	5.3	11.0	12.1	10.1	9.4
τ_p (s)	19.2	67.5	76.4	68.3	47.8
	0.21	0.25	0.63	0.40	0.13

interestingly, removing the chelator-soluble pectin leads to a greater decrease in the onion wall modulus, which is the result of this fraction being primarily present in a gel-like form and available to affect some degree of effective microfibril cross-linking as suggested by scattering experiments. $\bar{G}$ measures the fraction of structural strength that remains after polymer relaxation. As expected, this number is more clearly dependant on pectin concentration than τ_v , decreasing very significantly for the EDTA treated onion sample.

The viscoelastic contribution of different pectic fractions was also measured in SAOS measurements (Table 4 and Fig. S3). All cell wall materials showed soft gel behaviour with G' larger than G'' and $\tan \delta$ values between 0.1 and 0.3. Onion cell walls had a higher elastic modulus (G') than carrot and apple; this is likely to be related to the larger pectin content of the onion cell walls (Table 1). Carrot and apple cell walls had similar elastic moduli, and lower than onion cell wall, which could be due to their similar pectin content. Removal of acid extracted and chelating fractions led to a reduction in viscoelastic moduli confirming that all pectin fractions contribute to the elastic behaviour of cell walls.

4. Conclusions

Structural and material testing and analysis techniques previously applied to (bacterial) cellulose composites with cell wall polysaccharides are found to be suitable for the characterisation of plant cell wall materials. This includes the application of a core-shell model to represent the small angle scattering behaviour of cellulose in the presence of other polysaccharides, and the use of a poroviscoelastic model to characterise the mechanical properties of cell walls under compression. Scattering experiments show that, in the walls studied, pectin polymers are in sufficiently close proximity to cellulose to influence crystallinity and microfibril interactions. Under compression-relaxation testing, pectin and cellulose each have characteristic mechanical properties, with a poroviscoelastic model providing good fits. A key role for pectin in the cell walls studied is to control the properties of water at both a local level to influence molecular interactions and at a bulk scale to control fluid flow and consequent material properties. In addition, different microstructural and mechanical roles are inferred for galactan sidechains in influencing fluid flow, and calcium cross-linking in preventing association of pectin chains with cellulose. All of the features observed in both microstructural and mechanical properties of primary cell walls mirror those found in self-assembled cellulose composites of bacterial cellulose with pectins, although the extents of structural effects and timescales of some mechanical effects show interesting differences, deserving of further study.

Acknowledgements

The authors would like to thank Prof Jason R. Stokes for access to the rheometer. This study was funded by the Australian Research Council Centre of Excellence in Plant Cell Walls CE110001007.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodhyd.2019.105253>.

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